



American International University- Bangladesh (AIUB)

Faculty of Engineering (FE)

Course Name :	Data Communication Lab	Course Code :	COE 3201
Semester :	Spring 2025-26	Sec :	M
Faculty Name :	Md. Alomgir Kabir	Group :	02

Experiment No :	07
Experiment Name :	Study of Amplitude Modulator and Demodulator using Simulink

Submitted by (NAME):	MOABIA	Student ID:	23-55531-3
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Performance Date :	21/04/26	Submission Date :	05/05/26
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Marking Rubrics (to be filled by Faculty)

Category	Proficient [6]	Good [4]	Acceptable [2]	Unacceptable [1]	Secured Marks
Abstract, Objectives, Theoretical Background	All information, measures and variables are provided and explained.	All Information provided that is sufficient, but more explanation is needed.	Most information correct, but some information may be missing or inaccurate.	Much information missing and/or inaccurate.	
Results	All of the criteria are met; results are described clearly and accurately;	Most criteria are met, but there may be some lack of clarity and/or incorrect information.	Experimental results don't match exactly with the theoretical values and/or analysis is unclear.	Experimental results are missing or incorrect;	
Discussion	Demonstrates thorough and sophisticated understanding. Conclusions drawn are appropriate for analyses;	Hypotheses are clearly stated, but some concluding statements not supported by data or data not well integrated.	Some hypotheses missing or misstated; conclusions not supported by data.	Conclusions don't match hypotheses, not supported by data; no integration of data from different sources.	
General formatting	Title page, placement of figures and figure captions, and other formatting issues all correct.	Minor errors in formatting.	Major errors and/or missing information.	Not proper style in text.	
Writing & organization	Writing is strong and easy to understand; ideas are fully elaborated and connected; effective transitions between sentences; no typographic, spelling, or grammatical errors.	Writing is clear and easy to understand; ideas are connected; effective transitions between sentences; minor typographic, spelling, or grammatical errors.	Most of the required criteria are met, but some lack of clarity, typographic, spelling, or grammatical errors are present.	Very unclear, many errors.	
Comments:				Total Marks (Out of 6):	

Abstract:

This experiment is designed to-

- 1.To understand the use of Simulink for AM modulation.
- 2.To develop understanding of AM demodulation.

Problem Statement

In a communication system, message signals cannot be transmitted efficiently over long distances in their original low-frequency form because they suffer from high attenuation, noise, and poor radiation properties. Therefore, modulation is required to combine the message signal with a high-frequency carrier signal, which makes transmission more reliable and suitable for long-distance communication. In this experiment, both Amplitude Modulation (AM) and Frequency Modulation (FM) are implemented using Simulink, and demodulation techniques are applied to recover the original message signal from the modulated signal and analyze system performance.

Objectives:

1. To understand the concept of modulation and demodulation.
2. To implement Amplitude Modulation (AM) using Simulink.
3. To implement AM demodulation to recover the original signal.
4. To implement Frequency Modulation (FM) using Simulink.
5. To observe and analyze the output waveforms of modulated and demodulated signals.
6. To compare the performance of AM and FM techniques.

Background Study:

Modulation is the process of combining a message signal with a carrier signal for transmission. In Amplitude Modulation (AM), the amplitude of the carrier signal varies according to the message signal while the frequency remains constant. In Frequency Modulation (FM), the frequency of the carrier signal varies while the amplitude remains constant.

Demodulation is the reverse process where the original message signal is extracted from the modulated signal. AM demodulation can be done using envelope detection, while FM demodulation commonly uses Phase Locked Loop (PLL). These techniques are widely used in radio communication systems.

Problem Formulation:

AM Part:

Message signal:

$$m(t) = 2\sin(2\pi 4t) + 3\cos(2\pi 6t)$$

Carrier signal:

$$c(t) = \cos(2\pi 50t)$$

The message signal is modulated using AM technique in Simulink. Then demodulation is applied to recover the original signal.

FM Part:

Message signal:

$$m(t) = a \sin(2\pi f_m t + \pi/3)$$

Where,

$$a = 2, f_m = 10$$

The signal is modulated using FM and then demodulated using PLL-based system to retrieve the original signal.

Discussion:

The experiment shows how modulation helps in transmitting signals efficiently. In AM, the amplitude of the carrier changes with the message signal, which makes it simple but more sensitive to noise. In FM, the frequency changes instead of amplitude, which provides better noise resistance and signal quality.

From the Simulink results, it is observed that both AM and FM signals can be successfully demodulated to recover the original message signal. However, FM gives more stable and noise-resistant output compared to AM.

Reference:

1. Data Communication Laboratory Manual, AIUB.
2. Behrouz A. Forouzan, *Data Communications and Networking*.
3. MATLAB Simulink Documentation.

Results:

Part 01:

You have a signal ' $m(t) = (2*\sin(2*\pi*4*t)+3*\cos(2*\pi*6*t))$ '. Apply amplitude modulation (AM) on the given signal with carrier signal ' $c(t) = \cos(2*\pi*50*t)$ ', and then do demodulation to recover the original message signal $m(t)$.

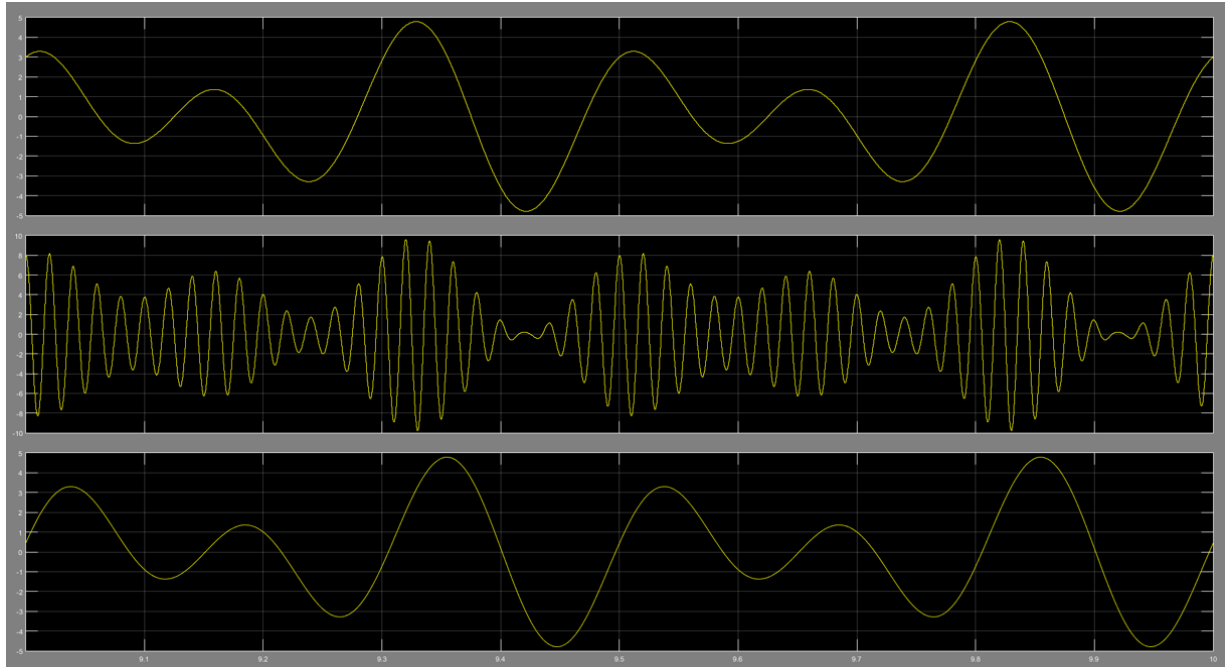


Figure 01: AM waveforms (Message signal, AM Modulated signal, Demodulated Signal)

Part 02 :

Message signal, $m(t) = a*\sin(2\pi f_m t + \pi/3)$, $a = 2$, $f_m = 10$. Use FM modulation and demodulation on the given signal and use two scopes to show your output. First scope should show message signal and modulated signal. Second scope should show message signal and demodulated signal.

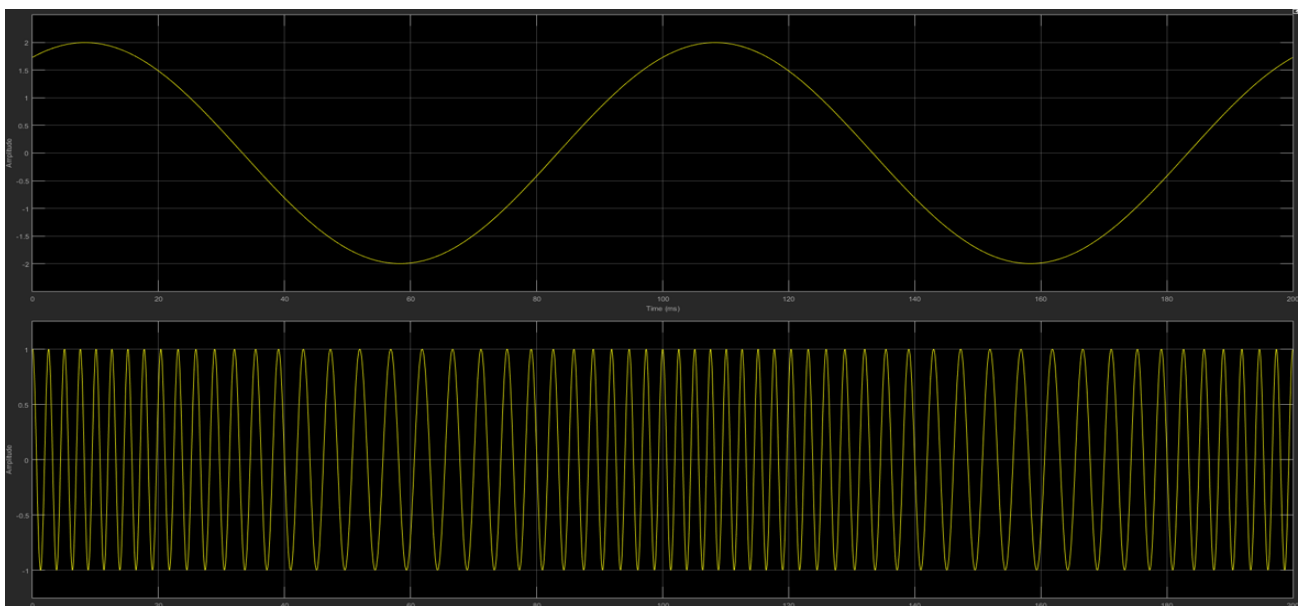


Figure 02: FM Message signal and FM Modulated signal

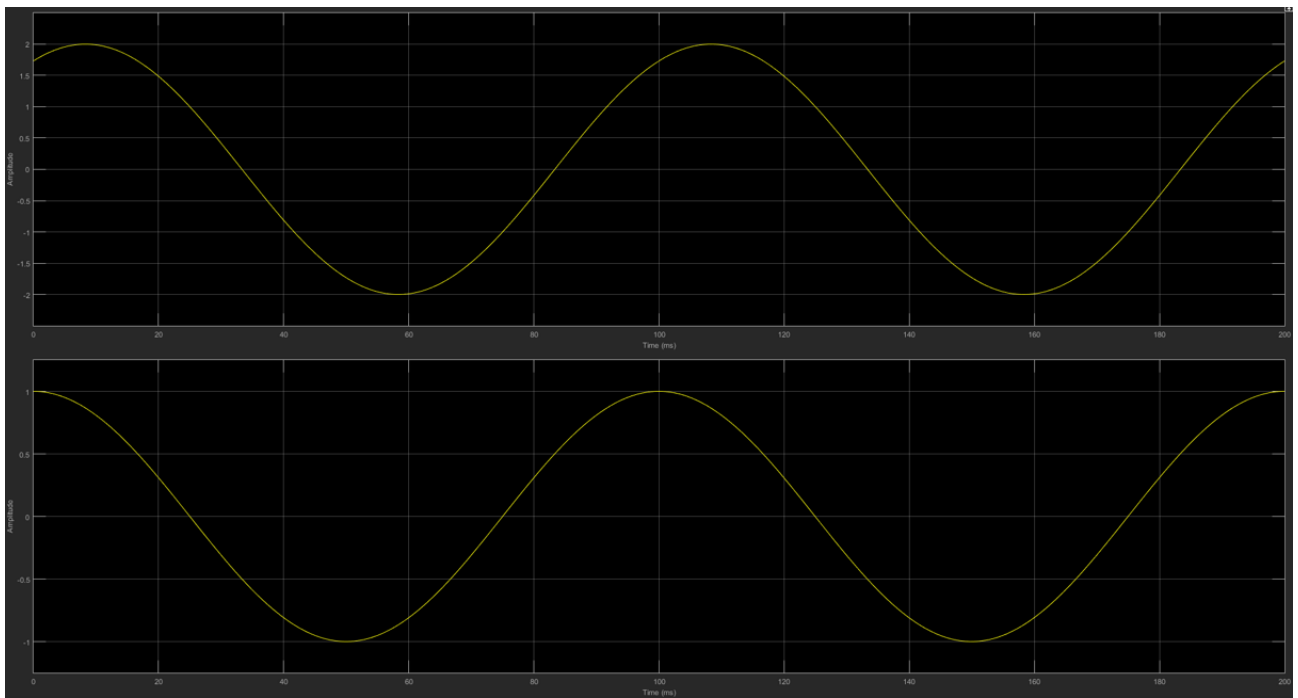


Figure 03: FM Message signal and FM Demodulated signal

Appendix

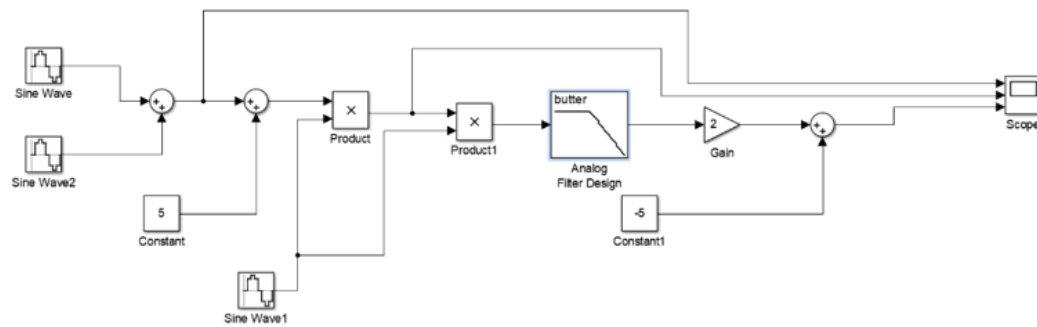


Figure 04: AM Modulator and AM Demodulator

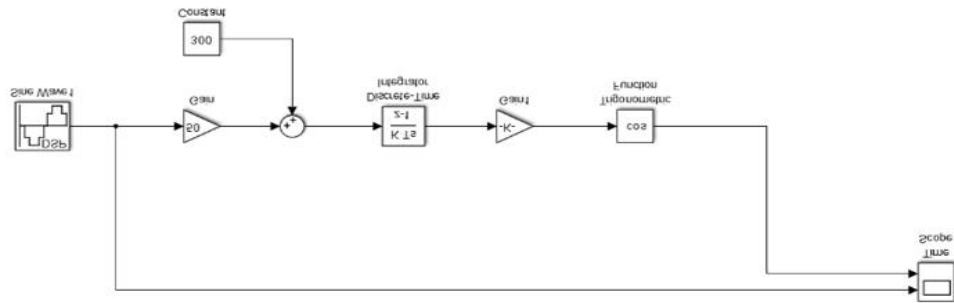


Figure 05: FM Modulator

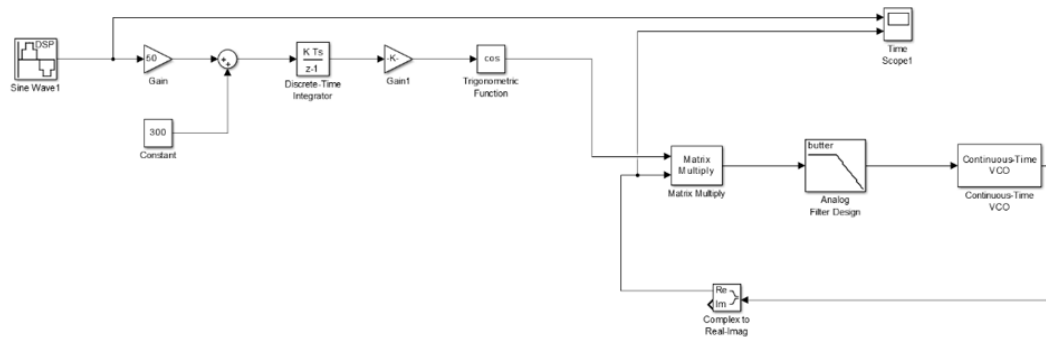


Figure 06: FM Demodulator

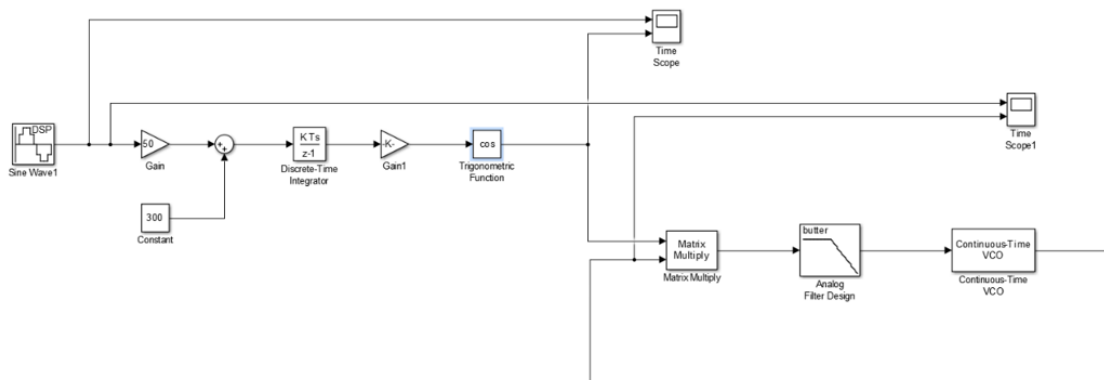


Figure 07: FM Modulator & Demodulator